Anaphylaxis/Allergic Reaction

A. Initial

a. Follow General Pre-Hospital Care-Treatment Protocol.

b. Pediatric patients (< 14 years of age) utilize MI MEDIC cards for appropriate medication dosage. When unavailable utilize pediatric dosing listed within protocol.

c. Ensure ALS response

d. Determine if anaphylaxis/severe allergic reaction (wheezing and/or hypotension) or an allergic reaction (itching, hives).

e. Determine substance or source of exposure, remove patient from source if known and able.

B. Anaphylaxis/Severe Allergic reaction

a. Assist patient in use of their own prescribed epinephrine auto-injector, if available

b. Administer epinephrine auto-Injector IM

1. Contact Medical Control if child appears to weigh less than 10 kg (approx. 20 lbs.), prior to epinephrine administration, if possible.

2. Administer pediatric epinephrine dose auto-injector IM if child weighs between 10-30 kg (approximately 20-60 lbs.)

3. Administer epinephrine auto-injector IM for adults and children weighing greater than 30 kg (approximately 60 lbs.)

4. May repeat epinephrine auto-injector IM one time after 3-5 minutes if the patient remains hypotensive, and auto-injector available

c. Administer epinephrine IM (per MCA selection may be BLS or MFR skill)

NOTE: BLS not carrying epinephrine auto-injector MUST participate in draw up epinephrine.

1. Contact Medical Control if child appears to weigh less than 10 kg (approx. 20 lbs.), prior to epinephrine administration, if possible.

2. Administer 0.15 mg (0.15 mL) of epinephrine IM (1mg/mL) if child weighs between 10-30 kg (approx. 20-60 lbs.)

3. Administer 0.3 mg (0.3 mL) of epinephrine IM (1mg/mL) for child weighing over 30 kg (approx. 60 lbs.) or adult patients.

4. May repeat epinephrine IM administration one time after 3-5 minutes if the patient remains hypotensive.

5. Maximum of 2 doses total of epinephrine (prescribed auto-injector, EMS supplied auto-injector, draw up epinephrine combined)

d. If wheezing and/or airway constriction, administer albuterol 2.5 mg/3mL NS nebulized (Per MCA selection may be EMT skill) per Medication Administration-Medication Protocol

1. If wheezing and/or airway constriction continues, administer nebulized albuterol 2.5 mg/3 ml NS nebulized and ipratropium 500 mcg/2.5 mL NS per Medication Administration-Medication Protocol (Per MCA selection may be Specialist skill)

e. For patients with hypotension administer NS or LR IV/IO fluid bolus (refer to Vascular Access and IV Fluid Therapy-Procedure Protocol) refer to Shock-Treatment Protocol.

1. Adults: up to 1 liter, wide open.

2. Pediatrics: 20 mL/kg, based on signs/symptoms of shock.

3. Fluid should be slowed to KVO when SBP greater than 90 mm/Hg.

f. Hypotensive patients should receive additional IV/IO fluid boluses, as indicated by hemodynamic state. (Consider preparing epi push dose while administering second bolus)

1. Adults: repeat IV/IO fluid bolus to a maximum of 2 liters.
2. Pediatrics: repeat dose of 20 mL/kg to a maximum of 40 ml/kg
3. Monitor for pulmonary edema.
4. If pulmonary edema presents, stop fluids and contact Medical Control for direction.

g. If hypotension persists/is unresponsive to fluid bolus, or severe respiratory distress is unresponsive to nebulized treatment, administer push dose epinephrine IV/IO.

Prepare (epinephrine 10 mcg/mL) by combining 1mL of 1mg/10mL epinephrine in 9mL NS

1. Adults:
   i. Administer 20 mcg (2 mL epinephrine 10 mcg/mL) IV/IO
   ii. Repeat every 3-5 minutes
   iii. Titrate SBP greater than 90 mm/Hg.
2. Pediatrics:
   i. Administer 1 mcg/kg (0.1 mL epinephrine 10 mcg/mL) IV/IO
   ii. Maximum dose 10 mcg (1 mL)
   iii. Repeat every 3-5 minutes

C. If patient is symptomatic of an allergic reaction but not in a severe allergic reaction or anaphylaxis OR after epinephrine administration:

a. Administer diphenhydramine.

1. Adult 50 mg IM or IV/IO
2. Pediatric 1 mg/kg IM/IV/IO (maximum dose 50 mg).

b. If wheezing, and albuterol not already administered, administer albuterol 2.5 mg/3mL NS nebulized (Per MCA selection may be EMT skill) per Medication Administration-Medication Protocol.

1. If wheezing continues, administer nebulized albuterol 2.5 mg/3 mL NS and ipratropium 500 mcg/2.5 mL NS per Medication Administration-Medication Protocol (Per MCA selection may be Specialist skill)

c. Administer prednisone tablet 50 mg PO to adults and children > 6 years of age (if available per MCA selection)

i. If prednisone is not available, patient is < 6 years of age, or patient is unable to receive medication PO, administer methylprednisolone IV/IO/IM:

a. Adults: 125 mg
b. Pediatrics: 2mg/kg (max 125 mg)

Appropriately trained EMT-Specialists (AEMT) may administer prednisone (PO) and methylprednisolone (IV/IO/IM) as specified in this protocol.

D. Patients unresponsive to treatment, contact Medical Control

MCA Approval of epinephrine auto-injector IM

☑ MFR

MCAs will be responsible for maintaining a roster of the agencies choosing to participate and will submit roster to MDHHS

MCA Approval of draw up epinephrine.

☑ MFR

☑ BLS

Personnel must complete MCA approved training prior to participating in draw up epinephrine.

MCAs will be responsible for maintaining a roster of the agencies choosing to participate and will submit roster to MDHHS.

Nebulized albuterol administration per MCA selection

☑ EMT

Nebulized albuterol/ipratropium administration per MCA selection

☑ Specialist

Nebulized albuterol administration per MCA Selection

☑ EMT

Nebulized albuterol/ipratropium administration per MCA selection

☑ Specialist

Additional Medication Option:

☑ Prednisone 50 mg tablet PO

(Adults and Children > 6 y/o)

Medication Protocols

Albuterol
Diphenhydramine
Epinephrine
Ipratropium
Methylprednisolone
Prednisone